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The evaluation and validation of Phadebas® paper as a presumptive screening tool for saliva on forensic exhibits

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ABSTRACT

The Phadebas® Forensic Press Test is routinely used for the detection of saliva. However, assessment of the use of Phadebas® paper for this purpose has not been studied extensively. The suitability of Phadebas® paper as a presumptive screening tool for saliva on forensic exhibits, was investigated by analysing the following: (1) sensitivity, (2) specificity, (3) effects of temperature on sensitivity and specificity, (4) detection of saliva in mixed body fluid samples, and (5) influence of substrate porosity. The results of this study demonstrated that Phadebas® paper is more sensitive to α-amylase activity and less specific for saliva than previously reported. The use of an examination temperature of 37 °C had no effect on sensitivity, but increased the incidence of cross-reactivity with other forensically relevant body fluid stains. Blood, urine and vaginal secretions can inhibit the detection of α-amylase activity with Phadebas® paper in mixed stains of saliva and body fluid. Substrate porosity is a weak predictor for the time taken for a saliva stain to achieve a strong positive result on Phadebas® paper. Overall, this study demonstrated that the Phadebas® Forensic Press Test has limitations as a presumptive test for the accurate identification of saliva.

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1. Introduction

Biological evidence originating from saliva can be important in criminal investigations; the detection and identification of saliva at a crime scene or on an exhibit may be crucial in establishing physical presence, or interaction, as well as inferring actions within a scene [1]. The forensic significance of saliva is often understated, yet the primary source of deoxyribonucleic acid (DNA) evidence recovered from volume crime scenes as reported in the United Kingdom arises from saliva, typically located on peripheral items such as drinking vessels and cigarette ends [2]. Furthermore, establishing the nature of, and involvement in a physical or sexual assault may rely on the detection of saliva on exhibits such as swabs, clothing or bedding from actions such as biting, saliva expectoration or oral intercourse [3,4]. Regardless of the means by which saliva becomes deposited at crime scenes or on exhibits, the practising forensic scientist is faced with one fundamental issue; the detection (or more specifically the localisation) of saliva.

Saliva stains may lack readily visible constituents and are therefore difficult to observe with the naked eye [5,6]. Consequently, screening techniques such as alternate light sources (ALS) or enzymatic screening techniques are required to enhance visualisation and detection. A 450 nm light source viewed through an orange filter is typically considered the most useful ALS method for observing saliva [7]. However, it is difficult to distinguish between stains originating from saliva and those from other body fluids, especially semen [5,7], and the fluorescence exhibited is typically weaker than for other body fluids [8–10]. For these reasons there is a requirement for a more sensitive and specific technique for the localisation and identification of saliva.

A number of enzymatic screening techniques, including Phadebas® paper (Magle Life Sciences, Lund, Sweden), have been developed for this purpose [11–14]. The preferred molecular target for the enzymatic detection of saliva is α-amylase (EC 3.2.1.1), due to its persistence and abundance in saliva stains [15]. In human saliva, α-amylase catalyses the random hydrolysis of α-1,4-glucosidic linkages in dietary starch, glycogen, and related poly- and oligo-saccharides [16]. Phadebas® chemistry exploits this activity of α-amylase; water-insoluble starch polymers are covalently bound to blue dye (CAS RN® 163062-28-0) molecules, which are liberated in the presence of α-amylase. For Phadebas® paper, these starch-dye complexes are immobilised onto filter paper and visualisation of α-amylase positive areas on exhibits occurs when the blue dye molecules diffuse through the pores of

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the filter paper. The major issue concerning the use of Phadebas® paper for the localisation of saliva is that α-amylase may also be present in other forensically relevant body fluids [17–22].

Studies on the effectiveness of Phadebas® paper as a presumptive screening tool for saliva are limited and require additional evaluation [3,11,23–27]. Detailed knowledge of the limitations of Phadebas® paper is crucial for the correct interpretation and inclusion of test results into forensic investigations. Therefore, this study aims to assess the suitability of Phadebas® paper as a presumptive screening tool for saliva on forensic exhibits. In order to achieve this, five criteria were identified for validation. Specifically, the sensitivity and specificity, and effects of temperature on the sensitivity and specificity of Phadebas® paper were evaluated. Additionally, the ability of Phadebas® paper to detect α-amylase activity in mixed saliva:body fluid stains was examined. Finally, the influence of substrate porosity on the ability of Phadebas® paper to detect saliva was analysed.

2. Materials and methods

2.1. Sample collection, storage and handling

After approval from the Human Research Ethics Committee (HREC) of Murdoch University (2017/040), human saliva (whole), blood (venous), faeces, nasal secretions, perspiration, semen, tear fluid, urine, and vaginal secretions were obtained from a total of five donors (two females and three males) aged between 22 and 60 years. Not all donors submitted each body fluid type for analysis and some submitted multiple samples of the same body fluid. To maximise variation in α-amylase activity, donors were instructed to provide samples at different times of the day. Saliva, perspiration, semen and urine were collected in sterile specimen containers, and faecal, nasal, tear and vaginal samples were collected using sterile rayon swabs. All samples (with the exception of blood) were immediately stored at −10 °C to preserve α-amylase activity [20,28]; blood was used immediately upon extraction. When required for use, frozen samples were thawed at room temperature (23 °C) and liquid samples were vortexed briefly. All frozen samples were used only once to avoid repeat freeze–thaw cycles.

2.2. Determination of sensitivity

2.2.1. Quantification of α-amylase activity

A stock solution of neat saliva was collected from a single donor and was separated into four specimen containers. One sample was used for the quantification of α-amylase activity in the stock saliva solution. Quantification was achieved (in triplicate) using the Phadebas® Amylase Test (Magle Life Sciences, Lund, Sweden) following the manufacturers' procedure for high α-amylase activities [29]. The absorbance of the supernatant was measured at 620 nm using a Hitachi U-1100 spectrophotometer.

2.2.2. Sample preparation for testing sensitivity

The second stock saliva sample was used to create 1:5, 1:10, 1:50, 1:100, 1:250, 1:500 and 1:1000 dilutions of saliva in distilled water in triplicate. Alpha-amylase activity was calculated for these dilutions based on the results from the Phadebas® Amylase Test method. Dilutions were stored on ice until required and vortexed briefly before use. A volume of 20 μL of each diluted sample was deposited onto glass sheets to maximise contact between the sample and Phadebas® paper. All glass sheets used were decontaminated with a 1:20 solution of TriGene and rinsed with distilled water prior to use. Fresh neat saliva and distilled water were used as the positive and negative control, respectively. Wet samples were examined at room temperature and in an incubator at 37 °C with Phadebas® paper, following the Phadebas® Forensic Press Test protocol [30]. This protocol was followed for all subsequent examinations; observations were recorded at 1, 2, 3, 4, 5, 10, 20, 30 and 40 min and were assigned a reaction score based on the colour intensity exhibited on the Phadebas® paper.

2.3. Determination of specificity

Saliva, blood, faecal, nasal, perspiration, semen, tear, urine, and vaginal samples were used to create stains by depositing 20 μL of liquid or swabbing the sample (where appropriate) onto glass sheets. Fresh neat saliva and distilled water were used as the positive and negative control, respectively. Stains were left to dry for 24 h at room temperature to simulate the state of stains (dried) encountered in forensic laboratories, before being examined with Phadebas® paper at room temperature and in an incubator at 37 °C.

2.4. Detection of saliva in mixed body fluid stains

The third stock saliva sample was used to create body fluid mixtures of saliva:blood, saliva:semen and saliva:urine in ratios of 1:0, 3:1, 1:1, 1:3 and 0:1, in triplicate. Samples were stored on ice until required and vortexed briefly before use. For each sample, 20 μL was deposited onto a glass sheet. Saliva:vaginal secretion stains were also generated by depositing 20, 10, 5, 2 and 1 μL of neat saliva from the saliva sample onto the glass sheet in duplicate. Vaginal secretion samples, that had been collected simultaneously, were subsequently swabbed over each wet saliva deposit. Fresh neat saliva and distilled water were used as the positive and negative control, respectively. Stains were left to dry for 24 h at room temperature before being examined with Phadebas® paper, at room temperature.

2.5. Detection of saliva on various substrates

2.5.1. Preparation of substrates

All textile substrates examined in this study originated from pre-owned garments acquired from an opportunity shop. Garment composition was as follows: garment 1, 100% neoprene; garment 2 and 5, 100% undetermined synthetic; garment 3 and 4, 100% polyester; garment 6, 100% silk; garment 7, 50% angora and 50% lambswool; garment 8–11, 100% cotton; garment 12, 85% cotton, 10% polyester and 5% elastane; garment 13, 60% nylon and 40% cotton; garment 14, 55% cotton and 45% polyester.

Garments were washed in a Bosch 7 kg front load washing machine as per the washing instruction label. Synthetic fibre (garment 1–5), natural fibre (garment 8–11) and blended fibre (garment 12–14) garments were washed on the 'Super15'' cycle with FAB® Fragrance Temptations™ laundry powder. This laundry powder is known to contain unspecified enzymes, however, it was not sampled as a negative control since α-amylase activity does not persist on textiles laundered under standard wash cycle conditions [25]. Delicate natural fibre (garment 6 and 7) garments were washed on the 'Delicate/Silk' cycle with Softly® Premium Laundry Liquid Delicates & Woollens. All garments were left to line-dry.

Paper samples were collected from an unopened quire or ream of sheets. Impermeable substrates were decontaminated with a 1:20 solution of TriGene and rinsed with distilled water, prior to use. Swatches measuring approximately 100 × 150 mm were excised, in duplicate, for all substrates and were stored at room temperature in separate paper bags until required.

2.5.2. Determination of substrate porosity by saturation

In order to determine substrate porosity (P_t), the total volume (V_t) of each substrate was initially determined. Each substrate was subsequently placed in a known volume of water; the

displacement of the water ($\Delta Volume_{H_2O}$) was then observed and substituted into Eq. (1) along with the total volume (V_t) of the substrate. This was repeated for all substrates.

$$\text{Porosity}\,(P_t)\% = (\text{Pore volume}\,(V_p)/\text{Total volume}\,(V_t)) \times 100 \quad (1)$$

where, V_t is equal to the L × W × H of the substrate, and V_p is equal to $V_t - \Delta Volume_{H_2O}$.

The porosity classification of the substrates in this study as defined by Eq. (1), were as follows: non-porous (0%), semi-porous (>0–10%), and porous (>10%).

2.5.3. Substrate stain preparation

A volume of 20 μL of the fourth stock saliva sample was deposited onto each of the various non-porous, semi-porous and porous substrates; this sample also acted as the positive control. Distilled water was used as the negative control. Stains were left to dry for 24 h at room temperature before being examined with Phadebas® paper, at room temperature.

3. Results

Since all observations of Phadebas® paper were manifested as qualitative results (i.e. colour intensity), interpretation was potentially influenced by examiner subjectivity. Thus, results were interpreted by a single examiner based on a devised qualitative scoring system for consistency (Fig. 1).

3.1. Sensitivity of Phadebas® paper

Following the quantification of α-amylase activity in the stock saliva sample (94,988 U/L ± 0.17%), dilutions produced from the solution were tested by observing the resulting colour reaction intensity on Phadebas® paper, in order to assess the limit of detection. Saliva samples with α-amylase activities of 94,988 U/L (± 0.17%) (neat) down to 189 U/L (1:500 dilution) generated a positive result on Phadebas® paper within the 40-min examination period at both room temperature and 37 °C (Table 1). The intensity of the colour reactions produced on Phadebas® paper for all three replicates of each dilution, at both temperatures were highly reproducible. The time to observe a positive result increased, as the dilution factor increased. Furthermore, the final colour intensity achieved at the end of the 40-min examination decreased as the dilution factor increased.

3.2. Specificity of Phadebas® paper

Saliva and other forensically relevant body fluid stains were tested by observing the intensity of the resulting colour reaction produced on Phadebas® paper, to assess cross-reactivity. The results obtained for saliva (n = 8), blood (n = 1), faecal (n = 8), nasal (n = 8), perspiration (n = 4), semen (n = 5), tear (n = 2), urine (n = 8) and vaginal (n = 2) samples with Phadebas® paper, at 23 °C versus 37 °C, are summarised in Table 2. Phadebas® paper was able to detect α-amylase activity in all saliva samples 1 min into the examination, at both temperatures; strong/very strong positive results were exhibited at 5 min for all saliva samples. Positive results were generated for faeces between 10 and 40 min at both temperatures. Nasal secretions produced mainly negative results, but a very weak result was observed at 40 min (23 °C), and a moderate result was produced at 20 min (37 °C). Similarly, perspiration, semen and urine exhibited mostly negative results, but produced a positive indication for α-amylase activity at 37 °C between 20 and 40 min. All blood, tear fluid and vaginal secretion samples resulted in negative observations.

To ensure that an adequate amount of biological material was transferred during sample deposition for samples collected using swabs, swabs were also dried for 24 h at room temperature and were directly examined with Phadebas® paper, at room temperature (data not shown). For this examination, 100% of faecal, nasal and tear fluid swabs returned positive results, observed after 10, 3 and 30 min into the examination, respectively. Vaginal swabs returned 100% negative results for α-amylase activity.

Blood, faeces, perspiration, semen and urine stains exhibited colour transfers to Phadebas® paper (Fig. 2). Blood developed an intense red-brown stain, while the positive reaction for α-amylase activity in faeces was typically accompanied by a green-brown stain. Perspiration displayed a white area, which was devoid of visibility of the blue starch polymer-dye complexes (as observed in the negative control). Semen was observed as a clear area, and urine transferred a faint yellow colouration to the Phadebas® paper.

3.3. Body fluid interference

To examine the possible interference of saliva detection in the presence of accompanying body fluids, mixed saliva:body fluid stains were tested by observing the resulting colour reaction intensity produced on Phadebas® paper. Results are summarised in Table 3. Despite being accompanied by a red-brown stain, α-amylase activity, in all stains consisting of saliva and blood, was detectable with Phadebas® paper. Compared to the positive control, the time to elicit any positive reaction was longer for saliva:blood stains at all ratios, especially the 1:3 stains, which only reached a moderate final colour intensity. The time required to achieve results of higher intensity increased as the ratio of saliva: blood decreased. This was similarly seen in the saliva:semen stains. Alpha-amylase activity in the saliva:semen stains was also detectable at all ratios with Phadebas® paper. Saliva in urine produced similar results to the positive control for the 3:1 and 1:1 stains; a negative result was obtained for the 1:3 saliva:urine stain. Due to this observation, a secondary examination for saliva:urine was undertaken with an alternative urine sample; all stains examined in this secondary test produced false negative results after the 40min examination with Phadebas® paper. Saliva:vaginal

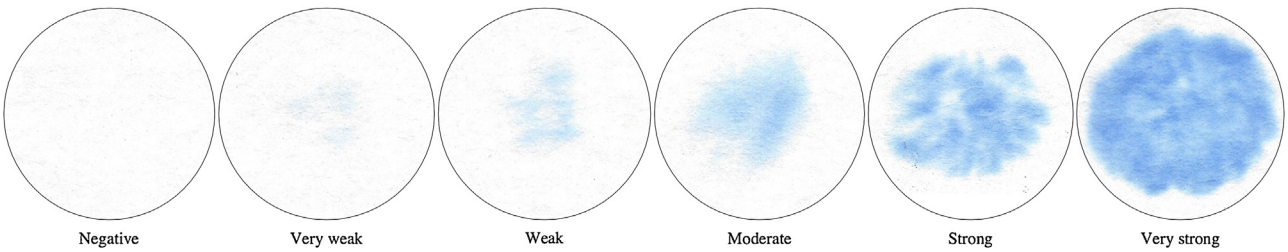


Fig. 1. Colour intensity scale for interpreting Phadebas® paper observations. Colour intensity scores were categorised as follows: negative (N); very weak (VW); weak (W); moderate (M); strong (S); very strong (VS). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 1
Phadebas® Forensic Press Test results for dilutions of liquid saliva examined at 23 °C and 37 °C.

Sample (n = 6)	α-amylase activity (U/L)	Time of observation (min)								
		1	2	3	4	5	10	20	30	40
Positive control	–	M	M	S	S	S	VS	VS	VS	VS
Negative control	–	N	N	N	N	N	N	N	N	N
Neat	94,988[a]	M	M	S	S	S	VS	VS	VS	VS
1:5	18,997	W	W	M	M	M	S	VS	VS	VS
1:10	9498	N	VW	W	M	M	M	VS	VS	VS
1:50	1899	N	N	N	VW	VW	W	M	M	M
1:100	949	N	N	N	N	N	VW	W	W	W
1:250	379	N	N	N	N	N	VW	VW	W	W
1:500	189	N	N	N	N	N	VW	VW	VW	VW
1:1000	94	N	N	N	N	N	N	N	N	N

Colour intensity score: N = negative; VW = very weak; W = weak; M = moderate; S = strong; VS = very strong.
–Samples of unknown α-amylase activity.
[a] Mean of three measurements, error margin ± 0.17%.

secretion mixtures displayed highly variable results. Due to a concern that an adequate amount of vaginal fluid was not mixed with the saliva deposits, swabs used to generate the saliva:vaginal secretion stains were allowed to dry for 24 h at room temperature and were directly examined with Phadebas® paper, at room temperature; similar highly variable results were observed (data not shown).

3.4. Influence of substrate porosity

To assess if substrate porosity influenced the detection of saliva, neat saliva stains deposited on various non-porous, semi-porous and porous substrates were tested by observing the intensity of the colour reaction produced on Phadebas® paper. Time to achieve a strong positive result on Phadebas® paper was plotted against porosity for each substrate to obtain a trend line equation ($y = 0.2117x + 6.381$) and a coefficient of determination (R^2) value (0.29798) (Fig. 3). The R^2 value indicated that the time required to achieve a strong positive result with Phadebas® paper was weakly dependent upon substrate porosity. In general, non-porous substrates achieved a strong positive result more quickly, compared to semi-porous and porous substrates. Positive indications for α-amylase activity were achieved in the 40-min examination period for all substrates tested. However, saliva stains on garment 11 and 12 failed to produce a strong positive final colour intensity with Phadebas® paper within the 40-min examination period, producing very weak and moderate results, respectively.

4. Discussion

The experiments undertaken in this study have identified certain limitations of Phadebas® paper as a presumptive screening tool for saliva during forensic examinations. In particular, the sensitivity, specificity, effects of temperature on sensitivity and specificity, ability to detect saliva (α-amylase activity) in mixed samples, and the effects of substrate porosity on the ability to detect saliva have been assessed effectively. Presumptive screenings tools are ideally, highly sensitive, specific and produce minimal false negative results [31]. This study has produced that suggest Phadebas® paper is more sensitive than what is currently reported; the manufacturer implies that the lower detection limit of Phadebas® paper for α-amylase activity is 2000 U/L [32]. However, it has been determined here that α-amylase activity of 189 U/L may be detected. Previous studies that have attempted to validate the sensitivity of Phadebas® paper have typically presented the lower detection limit as a dilution factor, neglecting any consideration for the initial α-amylase activity of the sample [3,11,23,26,27]. As a result, it has been reported that Phadebas® paper can detect α-amylase activity in saliva samples within a wide range of dilutions, between 1:100–1:1000; the value determined in this study (1:500) is within this range. This large variation in reported values may be explained by the existence of intra- and inter- personal variations of α-amylase activity in the population due to a number of factors, such as genetic influences [33], age [34], drug use [35,36], illness [37], diurnal rhythm [38], stress [39] and food intake [20]. An additional factor that may contribute to this variation is the decision to use dried/aged samples rather than fresh in such experiments; α-amylase activity has been shown to decrease to approximately 17% of the initial activity of the sample during a drying period of 24 h [20] and to 1% of the original α-amylase activity after 49 days [22]. By quantifying the α-amylase activity in the sample prior to examination and analysing a wet sample, this study has provided significance to the reported dilution factor and has negated any drying/ageing effects on α-amylase activity.

The specificity of Phadebas® paper is influenced by its sensitivity. The manufacturer states that, to assure selectivity for saliva, it must not detect stains with α-amylase activity below 2000 U/L in the 40-min examination. Many studies have previously identified that other body fluids including blood, breast milk, faeces, nasal secretions, perspiration, semen, tears and urine can contain α-amylase activities that exceed the lower sensitivity threshold determined in this study (approximately 189 U/L); based on the previously accepted value (2000 U/L), Phadebas® paper should, theoretically, only exhibit cross-reactivity with faeces, tears and urine [17–22]. This work has demonstrated that other forensically relevant body fluids additional to faeces, tear fluid and urine have the potential to exhibit cross-reactivity when examined with Phadebas® paper, including nasal secretions, perspiration and semen. In the literature there are conflicting findings surrounding the ability of other forensically relevant fluids to exhibit cross-reactivity with Phadebas® paper; typically, other forensically relevant body fluids do not exhibit cross-reactivty (with the exception of faeces) [3,40]. However, none of these studies examined nasal secretions or tear fluid. Although the sample size in this study was relatively small, similar results were obtained to that of Olsén et al. [24] and Davidek [26], which demonstrated cross-reactivity with perspiration, semen, urine and vaginal secretions. It should be noted that if this study were to be repeated, all samples should be collected in sterile specimen containers in preference to swabs, due to the difficulty encountered in transferring the samples collected with swabs onto the glass.

In this study, other forensically relevant body fluids such as blood, faeces, perspiration, semen and urine were typically

Table 2
Phadebas® Forensic Press Test results for saliva and other forensically relevant body fluids examined at 23 °C and 37 °C.

Sample	23 °C									37 °C								
	Time of observation (min)									Time of observation (min)								
	1	2	3	4	5	10	20	30	40	1	2	3	4	5	10	20	30	40
Control																		
Positive	M	M	M	S	S	VS	VS	VS	VS	M	M	S	S	S	VS	VS	VS	VS
Negative	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N
Saliva																		
1	M	M	M	S	S	VS	VS	VS	VS	S	S	S	VS	VS	VS	VS	VS	VS
2	W	W	M	S	S	VS	VS	VS	VS	W	W	M	M	S	VS	VS	VS	VS
3	W	M	S	S	S	S	VS	VS	VS	W	M	S	S	S	VS	VS	VS	VS
4	S	S	S	S	S	S	S	VS	VS	S	S	S	S	S	VS	VS	VS	VS
5	M	M	S	S	S	VS	VS	VS	VS	M	M	S	S	S	VS	VS	VS	VS
6	S	S	S	S	S	VS	VS	VS	VS	S	S	S	S	S	VS	VS	VS	VS
7, 8	M	S	S	S	S	VS	VS	VS	VS	M	S	S	VS	VS	VS	VS	VS	VS
Blood																		
1	N[a]	N[a]	N[a]	N[a]	N[a]	N[a]	N[a]	N[a]	N[a]	N[a]	N[a]	N[a]	N[a]	N[a]	N[a]	N[a]	N[a]	N[a]
Faeces																		
1	N[b]	N[b]	N[b]	N[b]	N[b]	VW[b]	W[b]	M[b]	S[b]	N	N	N	N	N	N	W	M	S
2	N[b]	N[b]	N[b]	N[b]	N[b]	VW[b]	VW[b]	VW[b]	W[b]	N[b]	N[b]	N[b]	N[b]	N[b]	N[b]	N[b]	N[b]	VW[b]
3	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N
4	N	N	N	N	N	N	N	VW	VW	N	N	N	N	N	N	VW	VW	M
5	N[b]	N[b]	N[b]	N[b]	N[b]	N[b]	N[b]	VW[b]	VW[b]	N[b]	N[b]	N[b]	N[b]	N[b]	N[b]	VW[b]	VW[b]	W[b]
6, 7	N[b]	N[b]	N[b]	N[b]	N[b]	VW[b]	VW[b]	M[b]	M[b]	N[b]	N[b]	N[b]	N[b]	N[b]	N[b]	W[b]	M[b]	S[b]
8	N[b]	N[b]	N[b]	N[b]	N[b]	N[b]	N[b]	VW[b]	W[b]	N[b]	N[b]	N[b]	N[b]	N[b]	N[b]	VW[b]	W[b]	S[b]
Nasal secretions																		
1, 2, 4–7	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N
3	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	M	M	M
8	N	N	N	N	N	N	N	N	VW	N	N	N	N	N	N	N	N	N
Perspiration																		
1	N[c]	N[c]	N[c]	N[c]	N[c]	N[c]	N[c]	N[c]	N[c]	N[c]	N[c]	N[c]	N[c]	N[c]	N[c]	N[c]	N[c]	N[c]
2	N[c]	N[c]	N[c]	N[c]	N[c]	N[c]	N[c]	N[c]	N[c]	N[c]	N[c]	N[c]	N[c]	N[c]	N[c]	N[c]	N[c]	VW[c]
3, 4	N[c]	N[c]	N[c]	N[c]	N[c]	N[c]	N[c]	N[c]	N[c]	N[c]	N[c]	N[c]	N[c]	N[c]	N[c]	VW[c]	W[c]	M[c]
Semen																		
1	N[d]	N[d]	N[d]	N[d]	N[d]	N[d]	N[d]	N[d]	N[d]	N[d]	N[d]	N[d]	N[d]	N[d]	N[d]	N[d]	VW[d]	W[d]
2	N[d]	N[d]	N[d]	N[d]	N[d]	N[d]	N[d]	N[d]	N[d]	N[d]	N[d]	N[d]	N[d]	N[d]	N[d]	N[d]	N[d]	N[d]
3, 5	N[d]	N[d]	N[d]	N[d]	N[d]	N[d]	N[d]	N[d]	N[d]	N[d]	N[d]	N[d]	N[d]	N[d]	N[d]	N[d]	N[d]	VW[d]
4	N[d]	N[d]	N[d]	N[d]	N[d]	N[d]	N[d]	N[d]	N[d]	N[d]	N[d]	N[d]	N[d]	N[d]	N[d]	N[d]	N[d]	W[d]
Tear fluid																		
1, 2	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N
Urine																		
1,2, 4–8	N[e]	N[e]	N[e]	N[e]	N[e]	N[e]	N[e]	N[e]	N[e]	N[e]	N[e]	N[e]	N[e]	N[e]	N[e]	N[e]	N[e]	N[e]
3	N[e]	N[e]	N[e]	N[e]	N[e]	N[e]	N[e]	N[e]	N[e]	N[e]	N[e]	N[e]	N[e]	N[e]	N[e]	N[e]	VW[e]	W[e]
Vaginal secretions																		
1, 2	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N

Colour intensity score: N = negative; VW = very weak; W = weak; M = moderate; S = strong; VS = very strong.
[a] Red-brown stain visible.
[b] Green-brown stain visible.
[c] White area visible.
[d] Clear area visible.
[e] Yellow stain visible.

associated with a colour transfer from the stain to the Phadebas® paper, allowing some degree of stain origin discrimination. However, at least one faecal sample and all nasal secretion samples reacted with Phadebas® paper without being accompanied by a colour transfer, akin to saliva stain interactions with Phadebas® paper. Thus, it is recommended that positive results on Phadebas® paper should not be assigned a class of origin based on the Phadebas® Forensic Press Test alone. To further enforce this recommendation, it has been shown in two studies [3,4], that only a small proportion (13%) of items that exhibit positive indications for saliva with Phadebas® paper, return positive results with the confirmatory test Rapid Stain Identification Saliva (RSID™-Saliva).

It is apparent from the sensitivity study that analysing saliva stains with Phadebas® paper at the optimum temperature for α-amylase activity (*i.e.* body temperature, 37 °C) does not increase sensitivity, which differs to results obtained by Hedman et al. [23]. This particular study demonstrated that the use of an incubator (at 37 °C) with Phadebas® paper resulted in a higher sensitivity of Phadebas® paper to the sample; a dilution of 1:200 could be detected at 37 °C, as opposed to 1:100 at room temperature, though with low reproducibility. However, in the specificity study of this work, examination at the higher temperature appeared to increase the incidence of cross-reactivity. This may be explained due to the use of dried samples as opposed to wet samples used in

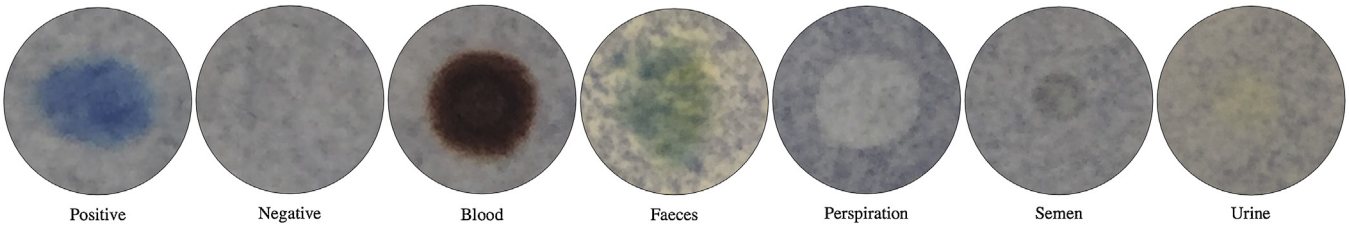


Fig. 2. Colour transferred from blood, faecal, perspiration, semen and urine stains on Phadebas® paper. Phadebas® paper exhibited red-brown staining in the presence of blood, green-brown staining in the presence of faeces, a white area in the presence of perspiration, a clear area in the presence of semen, and, yellow staining in the presence of urine. Positive and negative controls are also shown for reference. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 3
Phadebas® Forensic Press Test results for saliva:body fluid mixtures.

Sample	Time of observation (min)								
	1	2	3	4	5	10	20	30	40
Control (n = 3)									
Positive (1:0)	M	S	S	S	VS	VS	VS	VS	VS
Negative	N	N	N	N	N	N	N	N	N
Saliva:blood (n = 3)									
3:1	N[a]	N[a]	W[a]	W[a]	M[a]	S[a]	S[a]	VS[a]	VS[a]
1:1	N[a]	N[a]	VW[a]	VW[a]	W[a]	M[a]	S[a]	VS[a]	VS[a]
1:3	N[a]	N[a]	N[a]	VW[a]	VW[a]	W[a]	M[a]	M[a]	M[a]
0:1	N[a]	N[a]	N[a]	N[a]	N[a]	N[a]	N[a]	N[a]	N[a]
Saliva:semen (n = 3)									
3:1	W[b]	W[b]	M	S	VS	VS	VS	VS	VS
1:1	W	W	M	M	S	S	VS	VS	VS
1:3	VW	W	W	M	M	S	VS	VS	VS
0:1	N[b]	N[b]	N[b]	N[b]	N[b]	N[b]	N[b]	N[b]	N[b]
Saliva:urine* (n = 3)									
3:1	M	S	S	S	VS	VS	VS	VS	VS
1:1	W	M	S	S	VS	VS	VS	VS	VS
1:3	N	N	N	N	N	N	N	N	N
0:1	N	N	N	N	N	N	N	N	N
Saliva:vaginal secretions									
20 μL:swab 1	N	N	N	VW	VW	W	M	M	S
20 μL:swab 2	W	M	M	M	M	S	VS	VS	VS
10 μL:swab 3	N	VW	W	W	W	W	M	M	M
10 μL:swab 4	VW	W	W	W	W	M	M	S	S
5 μL:swab 5	N	N	N	N	N	VW	W	W	W
5 μL:swab 6	N	N	VW	VW	VW	W	M	M	S
2 μL:swab 7	N	N	N	N	VW	W	M	S	S
2 μL:swab 8	N	N	N	N	VW	VW	W	W	W
1 μL:swab 9	N	N	N	N	VW	VW	W	W	W
1 μL:swab 10	N	N	VW	VW	VW	W	M	M	M
Swab 11	N	N	N	N	N	N	N	N	N
Swab 12	N	N	N	N	N	N	N	N	N

Colour intensity score: N = negative; VW = very weak; W = weak; M = moderate; S = strong; VS = very strong.
* A secondary test using a different urine sample produced negative results for all ratios.
[a] Red-brown stain visible.
[b] Clear area visible.

the sensitivity study. Dried stains, in the presence of distilled water (used to moisten the exhibit and Phadebas® paper as per the current protocol), may take some time to become incorporated into a liquid again, which is required for α-amylase activity to be detected using Phadebas® paper. Due to an increase in kinetic energy of the molecules in the stains at a higher temperature (37 °C), the recovery of α-amylase activity may be achieved more rapidly, allowing the α-amylase in the stains to react with Phadebas® paper for a longer period of time. This may give a false indication of an increase in sensitivity and thus result in apparent enhanced cross-reactivity. It is known that some forensic laboratories have adopted changes in the Phadebas® Forensic Press Test protocol regarding the examination temperature by increasing it to 37 °C [41]. By implementing this change in examination temperature, there appears to be a trade-off between sensitivity and specificity. It is therefore recommended that to maintain specificity for saliva no modifications regarding examination temperature should be made to the current protocol.

This work found that for saliva in mixed body fluid stains, only semen did not adversely interfere with the detection of saliva with Phadebas® paper. In the samples where blood was the major component of the saliva:blood mixture, the overall observation on Phadebas® paper was substantially weaker than expected. Whole blood has been shown in one particular study to interfere with α-amylase activity in saliva; this was thought to be due to the high protein content [20], which may provide an explanation for the observations in this study. Variable reactions were observed with saliva in urine and vaginal secretion mixtures. It may be suggested that for these mixtures, the pH of the accompanying fluid may influence a decrease in α-amylase activity as a result of enzyme denaturation. It has previously been reported that the optimal pH for the activity of α-amylase (*in vitro*) is 7.0 [42]. The pH of blood is tightly regulated, between pH 7.35–7.45 [43]; the normal pH range of semen is 7.2–7.8 [44]; urine can widely range from pH 4.6–8.0 [45]; and vaginal secretions typically have a pH of 3.8–4.5 [46]. Further supporting this idea is despite urine possessing the next highest average amount of α-amylase activity, other than saliva or faeces [18], the urine samples examined in this study reacted less frequently when compared to semen, which is typically more neutral than urine.

The manufacturer states that no other forensically relevant body fluid (other than faeces) will react within 10 min following the current protocol [32]. However, no studies prior to this have investigated the potential influence of the substrate on the detectability of α-amylase activity with Phadebas® paper. This study demonstrated that the time to achieve a strong positive result varies greatly for saliva deposited on different substrates; porosity appeared to only be a weak predictor for this. Furthermore, this study has demonstrated that substrate composition may influence the detection of α-amylase activity with Phadebas® paper. For the most porous (approximately 84%) natural fibre (100% cotton) substrate tested in this work, Phadebas® paper failed to detect α-amylase activity until 40 min had elapsed and even then produced only a very weak result. Alpha-amylase activity was detectable in 1 min when the same saliva sample was deposited on the less-porous or non-porous substrates. Thus, it is possible that not even neat saliva will be detected by Phadebas® paper within 10 min, due to the influence of the substrate. Many previous studies investigating the sensitivity and specificity of Phadebas® paper deposit saliva samples onto cotton substrates without acknowledging the possible implications of doing so; the results obtained for these studies may be distorted due to the choice of substrate used for sample deposition [3,11,23,24,26,40]. It is recommended that

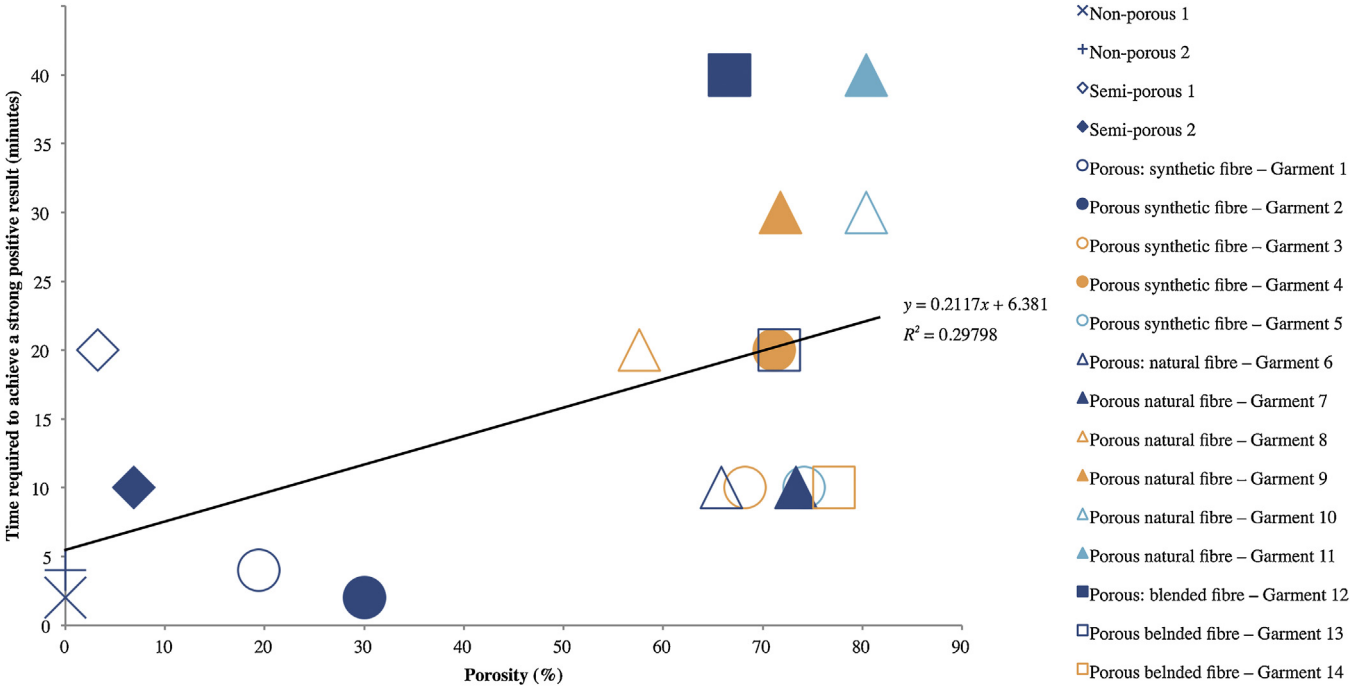


Fig. 3. Influence of substrate porosity (%) on the time required to achieve a strong positive reaction with Phadebas® paper. Garment 11 and 12 failed to achieve a strong final colour intensity within the 40-min examination period, very weak and moderate final colour intensites were observed, respectively, for these two substrates. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

future studies consider the potential effect of the substrate upon which saliva is deposited prior to attempting to define the limitations of Phadebas® paper.

5. Conclusion

Phadebas® paper is more sensitive to α-amylase activity and less specific for saliva (especially when using an examination temperature of 37 °C) than has previously been reported. Furthermore, mixed body fluid samples and porosity of the substrate may potentially influence the occurrence of false negative results. Therefore, Phadebas® paper does not possess the qualities required to accurately indicate the presence of saliva; the importance of utilising confirmatory testing, such as RSID™-Saliva following the Phadebas® Forensic Press Test, is reinforced by this study. Despite this, Phadebas® paper is useful as a screening tool and may rival the use of ALS techniques for the screening of saliva. In future studies, comparative analyses of Phadebas® paper and ALS should be conducted to determine the method that is most accurate and cost-effective, and least destructive to the sample, for use in forensic laboratories. The detectability of aged stains with Phadebas® paper also requires investigation.

Conflict of interest

The authors declare no conflicts of interest.

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